Self Learnign Assignment Day 1
C++ Programming

Name : **Abhinay Kumar**
PRN : **260240120005**
Form No : 251201864
Course : PGCP - AC
Batch : February 2026

1. Write a C++ program to calculate Simple and Compound interest

Program:

```
#include <iostream>
using namespace std;

// function to find power
float power(float base, int exp)
{
   float result = 1;

   for(int i = 1; i <= exp; i++)
   {
       result = result * base;
   }

   return result;
}

int main()
{
   float principal, rate, time;
   float SI, CI;

   cout << "Enter Principal Amount: ";
   cin >> principal;

   cout << "Enter Rate of Interest: ";
   cin >> rate;

   cout << "Enter Time (years): ";
   cin >> time;

   // Simple Interest
   SI = (principal * rate * time) / 100;
```

```
    // Compound Interest using power function
    CI = principal * power((1 + rate/100), time) - principal;

    cout << "\nSimple Interest = " << SI << endl;
    cout << "Compound Interest = " << CI << endl;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ intrest.cpp -o intre
/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"intrest
Enter Principal Amount: 10000
Enter Rate of Interest: 5
Enter Time (years): 3

Simple Interest = 1500
Compound Interest = 1576.25
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

2. Write a C++ Program to transpose one matrix and display both the matrices with proper prompt

Program:

```
#include <iostream>
using namespace std;

int main()
{
    int a[10][10], transpose[10][10];
    int rows, cols;

    cout << "Enter number of rows: ";
    cin >> rows;

    cout << "Enter number of columns: ";
    cin >> cols;

    cout << "Enter elements of matrix:\n";

    for(int i = 0; i < rows; i++)
    {
        for(int j = 0; j < cols; j++)
        {
            cin >> a[i][j];
```

```
        }
    }

    // Transpose of matrix
    for(int i = 0; i < rows; i++)
    {
        for(int j = 0; j < cols; j++)
        {
            transpose[j][i] = a[i][j];
        }
    }

    cout << "\nOriginal Matrix:\n";
    for(int i = 0; i < rows; i++)
    {
        for(int j = 0; j < cols; j++)
        {
            cout << a[i][j] << " ";
        }
        cout << endl;
    }

    cout << "\nTranspose Matrix:\n";
    for(int i = 0; i < cols; i++)
    {
        for(int j = 0; j < rows; j++)
        {
            cout << transpose[i][j] << " ";
        }
        cout << endl;
    }

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % cd "/Users/
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ transpose.cpp -o transpose && "
yaay/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"transpose
Enter number of rows: 3
Enter number of columns: 4
Enter elements of matrix:
10 20 30
20 40 50
20 40 50
50 68 95

Original Matrix:
10 20 30 20
40 50 20 40
50 50 68 95

Transpose Matrix:
10 40 50
20 50 50
30 20 68
20 40 95
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

3.Write a C++ program to process students' marks with the help of classes. The class has private variables for name, mark 1, mark 2, mark 3. It has two member functions - getdata() - to get input and result() - to print the results. All subjects mark must be >=50 for pass otherwise fail. Implement this program for n number of students.

Program:

```
#include <iostream>
using namespace std;

class Student
{
private:
   string name;
   int mark1, mark2, mark3;

public:
   // Function to get data
   void getdata()
   {
       cout << "Enter Student Name: ";
       cin >> name;

       cout << "Enter Marks of Subject 1: ";
       cin >> mark1;

       cout << "Enter Marks of Subject 2: ";
       cin >> mark2;

       cout << "Enter Marks of Subject 3: ";
```

```
        cin >> mark3;
    }

    // Function to display result
    void result()
    {
        cout << "\nStudent Name: " << name << endl;
        cout << "Marks: " << mark1 << " " << mark2 << " " << mark3 << endl;

        if(mark1 >= 50 && mark2 >= 50 && mark3 >= 50)
            cout << "Result: PASS\n";
        else
            cout << "Result: FAIL\n";
    }
};

int main()
{
    int n;

    cout << "Enter number of students: ";
    cin >> n;

    Student s[n];   // array of objects

    for(int i = 0; i < n; i++)
    {
        cout << "\nEnter details of student " << i+1 << endl;
        s[i].getdata();
    }

    cout << "\n--- Results ---\n";

    for(int i = 0; i < n; i++)
    {
        s[i].result();
    }

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % cd '
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ studentmarks.cpp -o stud
/abhinyaay/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"studentmarks
Enter number of students: 3

Enter details of student 1
Enter Student Name: abhishek
Enter Marks of Subject 1: 78
Enter Marks of Subject 2: 88
Enter Marks of Subject 3: 07

Enter details of student 2
Enter Student Name: ankit
Enter Marks of Subject 1: 67
Enter Marks of Subject 2: 487
Enter Marks of Subject 3: 9

Enter details of student 3
Enter Student Name: pawan
Enter Marks of Subject 1: 88
Enter Marks of Subject 2: 68
Enter Marks of Subject 3: 99

--- Results ---

Student Name: abhishek
Marks: 78 88 7
Result: FAIL

Student Name: ankit
Marks: 67 487 9
Result: FAIL

Student Name: pawan
Marks: 88 68 99
Result: PASS
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

4. Write a C++ program that allows the user to input 5 grades, i.e. marks between 0-100. The program must calculate the average mark, and state the number of marks less than 65.

Program:

```cpp
#include <iostream>
using namespace std;

int main()
{
   int marks[5];
   int sum = 0;
   int count = 0;

   cout << "Enter 5 marks (0-100):\n";

   for(int i = 0; i < 5; i++)
```

```
    {
        cin >> marks[i];
        sum += marks[i];

        if(marks[i] < 65)
        {
            count++;
        }
    }

    float average = sum / 5.0;

    cout << "\nAverage Marks = " << average << endl;
    cout << "Number of marks less than 65 = " << count << endl;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % cd "/
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ avgmark.cpp -o avgmark &&
/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"avgmark
Enter 5 marks (0-100):
88
89
78
99
67

Average Marks = 84.2
Number of marks less than 65 = 0
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

5. Write a menu driven C++ program to add, subtract and multiply given two matrices defined in class. Modify your program using operator overloading method

Program:

```
#include <iostream>
using namespace std;

class Matrix
{
private:
    int a[10][10], r, c;

public:
    void getData()
```

```
{
    cout << "Enter rows and columns: ";
    cin >> r >> c;

    cout << "Enter matrix elements:\n";
    for(int i=0;i<r;i++)
    {
        for(int j=0;j<c;j++)
        {
            cin >> a[i][j];
        }
    }
}

void display()
{
    for(int i=0;i<r;i++)
    {
        for(int j=0;j<c;j++)
        {
            cout << a[i][j] << " ";
        }
        cout << endl;
    }
}

// Addition
Matrix operator + (Matrix m)
{
    Matrix temp;
    temp.r = r;
    temp.c = c;

    for(int i=0;i<r;i++)
    {
        for(int j=0;j<c;j++)
        {
            temp.a[i][j] = a[i][j] + m.a[i][j];
        }
    }

    return temp;
}

// Subtraction
```

```
    Matrix operator - (Matrix m)
    {
        Matrix temp;
        temp.r = r;
        temp.c = c;

        for(int i=0;i<r;i++)
        {
            for(int j=0;j<c;j++)
            {
                temp.a[i][j] = a[i][j] - m.a[i][j];
            }
        }

        return temp;
    }

    // Multiplication
    Matrix operator * (Matrix m)
    {
        Matrix temp;
        temp.r = r;
        temp.c = m.c;

        for(int i=0;i<temp.r;i++)
        {
            for(int j=0;j<temp.c;j++)
            {
                temp.a[i][j] = 0;
                for(int k=0;k<c;k++)
                {
                    temp.a[i][j] += a[i][k] * m.a[k][j];
                }
            }
        }

        return temp;
    }
};

int main()
{
    Matrix m1, m2, result;
    int choice;
```

```
    cout << "Enter First Matrix\n";
    m1.getData();

    cout << "Enter Second Matrix\n";
    m2.getData();

    do
    {
        cout << "\n--- Matrix Menu ---\n";
        cout << "1. Addition\n";
        cout << "2. Subtraction\n";
        cout << "3. Multiplication\n";
        cout << "4. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch(choice)
        {
            case 1:
                result = m1 + m2;
                cout << "Addition Result:\n";
                result.display();
                break;

            case 2:
                result = m1 - m2;
                cout << "Subtraction Result:\n";
                result.display();
                break;

            case 3:
                result = m1 * m2;
                cout << "Multiplication Result:\n";
                result.display();
                break;
        }

    }while(choice != 4);

    return 0;
}
```

Output:

```
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ matric
s && "/Users/abhinyaay/Desktop/cpp /SLA_Day1_AbhinayKumar(2602
Enter First Matrix
Enter rows and columns: 2 2
Enter matrix elements:
10 20
30 40
Enter Second Matrix
Enter rows and columns: 3 3
Enter matrix elements:
20 40 50
20 20 10
40 70 90

--- Matrix Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Exit
Enter choice: 1
Addition Result:
30 60
50 60

--- Matrix Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Exit
Enter choice: 2
Subtraction Result:
-10 -20
10 20
```

```
--- Matrix Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Exit
Enter choice: 3
Multiplication Result:
600 800 700
1400 2000 1900

--- Matrix Menu ---
1. Addition
2. Subtraction
3. Multiplication
4. Exit
Enter choice: 4
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(26
```

6. Create a class DONOR that contains donor number, donor name, age address, sex and blood group. Write a Menu driven C++ program to display the number, name and address of the donors for the following categories:
a. blood donors having the blood group O+
b. blood donors in the age group between 18 to 25
c. female donors having blood group A in the age between 21 and 24.

Program:

```
#include <iostream>
using namespace std;

class DONOR
{
private:
   int donorNo, age;
   string name, address, sex, bloodGroup;

public:
   void getData()
   {
       cout << "Enter Donor Number: ";
       cin >> donorNo;

       cout << "Enter Name: ";
       cin >> name;

       cout << "Enter Age: ";
       cin >> age;

       cout << "Enter Address: ";
       cin >> address;

       cout << "Enter Sex (Male/Female): ";
       cin >> sex;

       cout << "Enter Blood Group: ";
       cin >> bloodGroup;
   }

   void display()
   {
       cout << donorNo << "  " << name << "  " << address << endl;
   }

   int getAge()
```

```
    {
        return age;
    }

    string getSex()
    {
        return sex;
    }

    string getBloodGroup()
    {
        return bloodGroup;
    }
};

int main()
{
    int n, choice;
    cout << "Enter number of donors: ";
    cin >> n;

    DONOR d[50];

    for(int i=0;i<n;i++)
    {
        cout << "\nEnter details of donor " << i+1 << endl;
        d[i].getData();
    }

    do
    {
        cout << "\nMENU\n";
        cout << "1. Donors with blood group O+\n";
        cout << "2. Donors age between 18 and 25\n";
        cout << "3. Female donors with blood group A (21-24)\n";
        cout << "4. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch(choice)
        {
            case 1:
                for(int i=0;i<n;i++)
                {
                    if(d[i].getBloodGroup()=="O+")
```

```
                    d[i].display();
            }
            break;

        case 2:
            for(int i=0;i<n;i++)
            {
                if(d[i].getAge()>=18 && d[i].getAge()<=25)
                    d[i].display();
            }
            break;

        case 3:
            for(int i=0;i<n;i++)
            {
                if(d[i].getSex()=="Female" &&
                   d[i].getBloodGroup()=="A" &&
                   d[i].getAge()>=21 && d[i].getAge()<=24)
                    d[i].display();
            }
            break;
    }

  }while(choice!=4);

  return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(26
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ tempCo
e && "/Users/abhinyaay/Desktop/cpp /SLA_Day1_AbhinayKumar(2602
Enter number of donors: 2

Enter details of donor 1
Enter Donor Number: 98237987
Enter Name: ankit
Enter Age: 22
Enter Address: agra
Enter Sex (Male/Female): male
Enter Blood Group: A+

Enter details of donor 2
Enter Donor Number: 3985
Enter Name: abhishek
Enter Age: 25
Enter Address: agra
Enter Sex (Male/Female): Male
Enter Blood Group: AB+

MENU
1. Donors with blood group O+
2. Donors age between 18 and 25
3. Female donors with blood group A (21-24)
4. Exit
Enter choice: 1

MENU
1. Donors with blood group O+
2. Donors age between 18 and 25
3. Female donors with blood group A (21-24)
4. Exit
Enter choice: 2
98237987  ankit  agra
3985  abhishek  agra
```

Ln 109, Col 2 (2163 selected) Spaces: 4

```
MENU
1. Donors with blood group O+
2. Donors age between 18 and 25
3. Female donors with blood group A (21-24)
4. Exit
Enter choice: 3

MENU
1. Donors with blood group O+
2. Donors age between 18 and 25
3. Female donors with blood group A (21-24)
4. Exit
Enter choice: 4
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(26
                                    Ln 109, Col 2 (2163 selected)    Spaces: 4
```

7. An electricity board charges the following rates to domestic users:
For the first 100 units - . 5.5P per unit
For next 200 units -. 7.25P per unit
Beyond 300 units - *. 8.75P per unit
All users are charged a minimum of₹. 60.
If the total amount is more than . 750 then an additional surcharge of 10% is added on the higher slab units. Write a C++ program to read the names of users and number of units consumed and print out the charges with names

Program:

```
#include <iostream>
using namespace std;

int main()
{
   string name;
   int units;
   float amount = 0, surcharge = 0, total;

   cout << "Enter User Name: ";
   cin >> name;

   cout << "Enter Units Consumed: ";
   cin >> units;

   // Calculate bill
   if(units <= 100)
   {
       amount = units * 5.5;
   }
```

```
    else if(units <= 300)
    {
        amount = (100 * 5.5) + ((units - 100) * 7.25);
    }
    else
    {
        amount = (100 * 5.5) + (200 * 7.25) + ((units - 300) * 8.75);
    }

    // Minimum charge
    if(amount < 60)
        amount = 60;

    // Surcharge
    if(amount > 750)
        surcharge = amount * 0.10;

    total = amount + surcharge;

    cout << "\nElectricity Bill\n";
    cout << "Name: " << name << endl;
    cout << "Units Consumed: " << units << endl;
    cout << "Amount: Rs. " << amount << endl;
    cout << "Surcharge: Rs. " << surcharge << endl;
    cout << "Total Bill: Rs. " << total << endl;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(26
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ lightb
yaay/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"lightbi
Enter User Name: abhinay
Enter Units Consumed: 250

Electricity Bill
Name: abhinay
Units Consumed: 250
Amount: Rs. 1637.5
Surcharge: Rs. 163.75
Total Bill: Rs. 1801.25
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(26
                                    Ln 48, Col 2 (967 selected)    Spaces: 4
```

8. Implement the calculator through a C++ program (use switch). The program should ask the user to input two operands and an operator. Your program should print the result of performing the required operation on the input operands as the following output:
Enter operand 1:5
Enter operand 2:10
Enter operation: /
The result of 5/10 is 0.5

Program:

```
#include <iostream>
using namespace std;

int main()
{
   float operand1, operand2, result;
   char op;

   cout << "Enter operand 1: ";
   cin >> operand1;

   cout << "Enter operand 2: ";
   cin >> operand2;

   cout << "Enter operation (+, -, *, /): ";
   cin >> op;

   switch(op)
   {
       case '+':
           result = operand1 + operand2;
           break;

       case '-':
           result = operand1 - operand2;
           break;

       case '*':
           result = operand1 * operand2;
           break;

       case '/':
           if(operand2 != 0)
               result = operand1 / operand2;
           else
           {
```

```
                cout << "Division by zero not allowed";
                return 0;
            }
            break;

        default:
            cout << "Invalid operator";
            return 0;
    }

    cout << "The result of " << operand1 << op << operand2 << " is " << result;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ calculator.cpp -o ca
inyaay/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"calculator
Enter operand 1: 20
Enter operand 2: 10
Enter operation (+, -, *, /): +
The result of 20+10 is 30%
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
                                        Ln 50, Col 2   Spaces: 4   UTF-8   LF   {
```

9. Create Time Class which contains the members as hour, minute and second. Write a C++ program that accepts member data for one Time class object from the keyboard. Then, ask the user to perform one of the following functions to that Time object:

- **AddHours**
- **AddMinutes**
- **AddSeconds**

And finally display the newly modified Time information.

Program:

```
#include <iostream>
using namespace std;

class Time
{
private:
    int hour, minute, second;

public:
    void getTime()
    {
```

```
        cout << "Enter Hours: ";
        cin >> hour;

        cout << "Enter Minutes: ";
        cin >> minute;

        cout << "Enter Seconds: ";
        cin >> second;
    }

    void addHours(int h)
    {
        hour += h;
    }

    void addMinutes(int m)
    {
        minute += m;
        hour += minute / 60;
        minute = minute % 60;
    }

    void addSeconds(int s)
    {
        second += s;
        minute += second / 60;
        second = second % 60;
        hour += minute / 60;
        minute = minute % 60;
    }

    void display()
    {
        cout << "Modified Time: ";
        cout << hour << ":" << minute << ":" << second << endl;
    }
};

int main()
{
    Time t;
    int choice, value;

    t.getTime();
```

```
    cout << "\n1. Add Hours\n";
    cout << "2. Add Minutes\n";
    cout << "3. Add Seconds\n";
    cout << "Enter choice: ";
    cin >> choice;

    cout << "Enter value to add: ";
    cin >> value;

    switch(choice)
    {
        case 1:
            t.addHours(value);
            break;

        case 2:
            t.addMinutes(value);
            break;

        case 3:
            t.addSeconds(value);
            break;

        default:
            cout << "Invalid choice\n";
    }

    t.display();

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % cd
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ time.cpp -o time && "/U
op/cpp /SLA_Day1_AbhinayKumar(260240120005)/"time
Enter Hours: 10
Enter Minutes: 15
Enter Seconds: 20

1. Add Hours
2. Add Minutes
3. Add Seconds
Enter choice: 1
Enter value to add: 1
Modified Time: 11:15:20
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

10. Write a menu driven program to perform the following options using separate functions. Accept a number and then, do the following operations on this number:
1.Prime number or not
2.Odd or Even
3.Positive or Negative
4.Exit

Program:

```
#include <iostream>
using namespace std;

// Function to check Prime
void checkPrime(int n)
{
   int count = 0;

   for(int i = 1; i <= n; i++)
   {
       if(n % i == 0)
           count++;
   }

   if(count == 2)
       cout << "Number is Prime\n";
   else
       cout << "Number is Not Prime\n";
}

// Function to check Odd or Even
void checkOddEven(int n)
{
   if(n % 2 == 0)
       cout << "Number is Even\n";
   else
       cout << "Number is Odd\n";
}

// Function to check Positive or Negative
void checkSign(int n)
{
   if(n > 0)
       cout << "Number is Positive\n";
   else if(n < 0)
       cout << "Number is Negative\n";
   else
       cout << "Number is Zero\n";
}
```

```
int main()
{
   int num, choice;

   cout << "Enter a number: ";
   cin >> num;

   do
   {
       cout << "\nMENU\n";
       cout << "1. Prime Number or Not\n";
       cout << "2. Odd or Even\n";
       cout << "3. Positive or Negative\n";
       cout << "4. Exit\n";
       cout << "Enter choice: ";
       cin >> choice;

       switch(choice)
       {
           case 1:
               checkPrime(num);
               break;

           case 2:
               checkOddEven(num);
               break;

           case 3:
               checkSign(num);
               break;

           case 4:
               cout << "Exiting program\n";
               break;

           default:
               cout << "Invalid choice\n";
       }

   }while(choice != 4);

   return 0;
}
```

Output:

```
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ checkN
ay/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"checkNum
Enter a number: 20

MENU
1. Prime Number or Not
2. Odd or Even
3. Positive or Negative
4. Exit
Enter choice: 1
Number is Not Prime

MENU
1. Prime Number or Not
2. Odd or Even
3. Positive or Negative
4. Exit
Enter choice: 2
Number is Even

MENU
1. Prime Number or Not
2. Odd or Even
3. Positive or Negative
4. Exit
Enter choice: 3
Number is Positive

MENU
1. Prime Number or Not
2. Odd or Even
3. Positive or Negative
4. Exit
Enter choice: 4
Exiting program
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(26
                                        Ln 82, Col 14    Spaces: 4
```

11. Write a function reverse() that reverses the character strings. Use it to write a program that reverses its input a line at a time.

Program:

```
#include <iostream>
#include <string>
using namespace std;

// Function to reverse string
```

```
string reverseStr(string str)
{
   string rev = "";

   for(int i = str.length() - 1; i >= 0; i--)
   {
       rev += str[i];
   }

   return rev;
}

int main()
{
   string line;

   cout << "Enter a line: ";
   getline(cin, line);

   cout << "Reversed line: " << reverseStr(line);

   return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ reverse.cpp -o r
/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"reverse
Enter a line: abhinay
Reversed line:  yanihba%
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ reverse.cpp -o r
/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"reverse
Enter a line: hello world
Reversed line: dlrow olleh%
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005
                                                Ln 28, Col 2   Spaces: 4   UTF-8
```

12. Write a C++ Program to Compare Two given Strings and display the result.

Program:

```
#include <iostream>
#include <string>
using namespace std;

int main()
{
   string str1, str2;
```

```
    cout << "Enter first string: ";
    cin >> str1;

    cout << "Enter second string: ";
    cin >> str2;

    if(str1 == str2)
        cout << "Both strings are equal";
    else
        cout << "Strings are not equal";

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % cd
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ compareString.cpp -o co
rs/abhinyaay/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"compareString
Enter first string: hello
Enter second string:  world
Strings are not equal%
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

13. Write a program in C++ to determine how many of the characters are vowels and how many are consonants in a given line of text. Also terminate the string when the input character encountered is other than the alphabets (a-z or A-Z) and Blank spaces.
Hint:
a. When the input string is 'C plus plus FOR BEGINNERS, TEST YOUR C plus plus PROGRAMMING STRENGTHS'. Program will consider the string 'C plus plus FOR BEGINNERS' only because ',' is encountered.
b. When the input string is 'C++ LEARINING IS NOT A PROBLEM'. Program will consider the character 'C' only because the '+' is encountered.

Program:

```
#include <iostream>
using namespace std;

int main()
{
    char str[200];
    int vowels = 0, consonants = 0;

    cout << "Enter a line of text: ";
    cin.getline(str, 200);
```

```cpp
    for(int i = 0; str[i] != '\0'; i++)
    {
        char ch = str[i];

        // Stop if character is not alphabet or space
        if(!((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z') || ch == ' '))
            break;

        if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
           ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U')
        {
            vowels++;
        }
        else if((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
        {
            consonants++;
        }
    }

    cout << "Number of vowels: " << vowels << endl;
    cout << "Number of consonants: " << consonants << endl;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % cd "/User
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ checkString.cpp -o checkStrin
bhinyaay/Desktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"checkString
Enter a line of text: hello world is THIS is a test
Number of vowels: 8
Number of consonants: 15
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

14. Create a C++ class for a stock item abstract data type. It should have the attributes of stock levels (an integer) and unit price (a float). Define the methods to return the values of these two attributes and to set them using parameters. Add two more methods to allow stock receipt and issues updating the stocklevel as appropriate. Write a menu driven C++ program to solve the problem.

Program:

```cpp
#include <iostream>
using namespace std;

class StockItem
{
private:
```

```
    int stockLevel;
    float unitPrice;

public:
    // Set values
    void setValues(int s, float p)
    {
        stockLevel = s;
        unitPrice = p;
    }

    // Get stock level
    int getStockLevel()
    {
        return stockLevel;
    }

    // Get unit price
    float getUnitPrice()
    {
        return unitPrice;
    }

    // Receive stock
    void receiveStock(int qty)
    {
        stockLevel += qty;
        cout << "Stock received successfully.\n";
    }

    // Issue stock
    void issueStock(int qty)
    {
        if(qty <= stockLevel)
        {
            stockLevel -= qty;
            cout << "Stock issued successfully.\n";
        }
        else
        {
            cout << "Not enough stock available.\n";
        }
    }
};
```

```
int main()
{
   StockItem item;
   int choice, stock, qty;
   float price;

   cout << "Enter Initial Stock Level: ";
   cin >> stock;

   cout << "Enter Unit Price: ";
   cin >> price;

   item.setValues(stock, price);

   do
   {
       cout << "\n---- MENU ----\n";
       cout << "1. Display Stock Details\n";
       cout << "2. Receive Stock\n";
       cout << "3. Issue Stock\n";
       cout << "4. Exit\n";
       cout << "Enter choice: ";
       cin >> choice;

       switch(choice)
       {
           case 1:
               cout << "Stock Level: " << item.getStockLevel() << endl;
               cout << "Unit Price: " << item.getUnitPrice() << endl;
               break;

           case 2:
               cout << "Enter quantity to receive: ";
               cin >> qty;
               item.receiveStock(qty);
               break;

           case 3:
               cout << "Enter quantity to issue: ";
               cin >> qty;
               item.issueStock(qty);
               break;

           case 4:
               cout << "Exiting program\n";
```

```
                break;

            default:
                cout << "Invalid choice\n";
        }

    } while(choice != 4);

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_Abhina
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g
e && "/Users/abhinyaay/Desktop/cpp /SLA_Day1_Abhinayk
Enter Initial Stock Level: 10000
Enter Unit Price: 4

---- MENU ----
1. Display Stock Details
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 1
Stock Level: 10000
Unit Price: 4

---- MENU ----
1. Display Stock Details
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 2
Enter quantity to receive: 3
Stock received successfully.

---- MENU ----
1. Display Stock Details
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 3
Enter quantity to issue: 1
Stock issued successfully.
```

```
---- MENU ----
1. Display Stock Details
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 2
Enter quantity to receive: 1
Stock received successfully.

---- MENU ----
1. Display Stock Details
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 1
Stock Level: 10003
Unit Price: 4

---- MENU ----
1. Display Stock Details
2. Receive Stock
3. Issue Stock
4. Exit
Enter choice: 4
Exiting program
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_Abhin
```

15. Raising a number n to power p is the same as multiplying by itself t times. Write a function called power() that makes a double value for n and an int value for p and returns the result as double value. Use default argument of 2 for p, so that if this argument is omitted, the number will be squared. Write a main() function that gets values from the user to test this function.

Program:

```
#include <iostream>
using namespace std;

// Function with default argument
double power(double n, int p = 2)
{
   double result = 1;
```

```
    for(int i = 1; i <= p; i++)
    {
        result = result * n;
    }

    return result;
}

int main()
{
    double n;
    int p;

    cout << "Enter number: ";
    cin >> n;

    cout << "Enter power (enter 0 to use default square): ";
    cin >> p;

    if(p == 0)
        cout << "Result = " << power(n) << endl;
    else
        cout << "Result = " << power(n, p) << endl;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % cd
sktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/" && g++ power.cpp -o power && "
ktop/cpp /SLA_Day1_AbhinayKumar(260240120005)/"power
Enter number: 10
Enter power (enter 0 to use default square): 3
Result = 1000
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```